## Zimbabwe: Debt Sustainability Analysis[24]

**Under the staff's baseline scenario, Zimbabwe's external debt is not expected to be sustainable in the absence of a debt rescheduling**. The baseline scenario assumes an average negative real GDP growth at 4 percent until 2010 and an average positive growth of 3 percent thereafter with exports growing at 5 percent. On the fiscal front, budgetary consolidation is not expected in the near future although deficit declines after 2009.

**Zimbabwe's total external public debt was US$5 billion (67 percent of GDP) at end-2004**, of which US$2 billion were owed to multilateral creditors. Domestic public debt denominated in local currency accounted for about 7 percent of GDP in 2004. The baseline assumes a tentative arrears rescheduling in 2011—outside of the medium-term scenario. The total stock of outstanding debt at the end of 2011 would amount to US$2.4 billion (34 percent of GDP).

**In the baseline scenario, all public and publicly guaranteed external debt indicators are projected to improve only slowly following the debt rescheduling.** The NPV of the debt-to-GDP ratios are expected to fall to about 50 percent, the NPV of the debt-to-exports ratios to fall about 130 percent and the debt service-to-exports would decline to 1 percent in the long-run. However, private external debt is projected to increase from 2010 given the need to obtain foreign financing.

**The debt dynamics are most vulnerable to a depreciation and export decline**. A one-time 30 percent depreciation would push up the NPV of the debt-to-GDP ratio to about 120 percent in 2010. A one-time two standard deviation shock in exports would raise the debt service-to-exports to 30 percent compared with 20 percent in 2025 in the baseline scenario, and the NPV of the debt-to-exports ratio would rise to 217 percent compared with 122 percent in the baseline scenario in 2015.

**The public debt sustainability analysis could not be conducted due to difficulty in choosing applicable exchange rates since the analysis is undertaken in national currency**. Given the highly overvalued official exchange rate and the multiple exchange rate system, large swings in the exchange rate have a dramatic effect on dollars amounts converted into national currency that are to be divided by GDP in national currency. To circumvent this problem we use nominal U.S. dollar GDP calculated by using the national real growth and the international inflation (1996 base year) in the external DSA since the analysis is undertaken in U.S. dollars.

[24] Standard Fund templates for low income countries and sensitivity tests are used to assess Zimbabwe's debt dynamics. The DSA was undertaken by the Fund only.

Table 1a. Zimbabwe: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

| | Actual | | | Historical Average 6/ | Standard Deviation 6/ | Estimate | Projections | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2002 | 2003 | 2004 | | | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2005-10 Average | 2015 | 2025 | 2011-25 Average |
| **External debt (nominal) 1/** | **53.0** | **64.4** | **67.4** | | | **74.4** | **80.6** | **85.9** | **91.0** | **95.8** | **100.5** | | **33.0** | **28.1** | |
| o/w public and publicly guaranteed (PPG) | 49.8 | 62.2 | 66.0 | | | 73.3 | 77.1 | 80.0 | 82.7 | 85.0 | 87.6 | | 22.5 | 21.0 | |
| Change in external debt | 7.7 | 11.4 | 3.1 | | | 7.0 | 6.2 | 5.4 | 5.0 | 4.8 | 4.8 | | -0.5 | -0.6 | |
| Identified net debt-creating flows | 3.7 | 9.6 | 7.1 | | | 12.6 | 7.5 | 7.4 | 7.2 | 7.2 | 7.4 | | 4.7 | 3.2 | |
| **Non-interest current account deficit** | **1.0** | **2.9** | **4.3** | **1.4** | **2.7** | **6.2** | **1.2** | **1.6** | **1.5** | **1.5** | **2.9** | | **5.9** | **4.6** | 5.5 |
| Deficit in balance of goods and services | 2.4 | 4.3 | 5.7 | | | 7.6 | 3.0 | 3.6 | 3.7 | 3.9 | 5.2 | | 4.2 | 2.3 | |
| Exports | 24.5 | 24.7 | 27.1 | | | 28.0 | 30.7 | 32.4 | 34.6 | 36.5 | 38.2 | | 38.2 | 40.0 | |
| Imports | 26.9 | 29.0 | 32.8 | | | 35.6 | 33.7 | 36.0 | 38.2 | 40.4 | 43.4 | | 42.4 | 42.4 | |
| Net current transfers (negative = inflow) | -2.8 | -2.2 | -2.8 | -2.0 | 0.5 | -2.7 | -2.8 | -2.9 | -2.9 | -2.9 | -2.9 | | -2.5 | -1.9 | -2.3 |
| Other current account flows (negative = net inflow) | 1.3 | 0.9 | 1.4 | | | 1.3 | 1.1 | 0.9 | 0.7 | 0.6 | 0.6 | | 4.2 | 4.2 | |
| **Net FDI (negative = inflow)** | **-0.3** | **0.0** | **-0.1** | **-0.9** | **1.4** | **-0.2** | **0.0** | **0.0** | **0.0** | **0.0** | **0.0** | | **-1.0** | **-1.0** | -1.0 |
| **Endogenous debt dynamics 2/** | **3.0** | **6.7** | **2.9** | | | **6.5** | **6.2** | **5.8** | **5.7** | **5.6** | **4.5** | | **-0.2** | **-0.4** | |
| Contribution from nominal interest rate | 1.6 | 1.7 | 1.4 | | | 1.4 | 2.6 | 2.9 | 3.1 | 3.4 | 3.7 | | 0.7 | 0.4 | |
| Contribution from real GDP growth | 2.0 | 6.0 | 2.7 | | | 5.1 | 3.6 | 2.9 | 2.5 | 2.2 | 0.7 | | -1.0 | -0.8 | |
| Contribution from price and exchange rate changes | -0.7 | -1.0 | -1.3 | | | … | … | … | … | … | … | | … | … | |
| **Residual (3-4) 3/** | **4.1** | **1.7** | **-4.0** | | | **-5.6** | **-1.3** | **-2.0** | **-2.1** | **-2.4** | **-2.6** | | **-5.2** | **-3.8** | |
| o/w exceptional financing | -6.9 | -5.9 | -3.3 | | | -5.1 | -4.3 | -4.6 | -4.7 | -4.6 | -5.8 | | 0.0 | 0.0 | |
| NPV of external debt 4/ | ... | ... | 68.8 | | | 75.1 | 80.8 | 85.9 | 90.8 | 95.6 | 100.5 | | 57.2 | 60.1 | |
| In percent of exports | ... | ... | 253.7 | | | 267.9 | 263.4 | 264.9 | 262.8 | 261.6 | 263.1 | | 149.9 | 150.0 | |
| **NPV of PPG external debt** | **...** | **...** | **67.4** | | | **74.0** | **77.3** | **80.0** | **82.5** | **84.9** | **87.5** | | **46.7** | **53.0** | |
| In percent of exports | ... | ... | 248.5 | | | 263.9 | 252.0 | 246.7 | 238.7 | 232.3 | 229.3 | | 122.4 | 132.3 | |
| Debt service-to-exports ratio (in percent) | 25.3 | 23.5 | 18.7 | | | 18.5 | 18.0 | 17.6 | 16.9 | 16.0 | 16.0 | | 5.4 | 2.7 | |
| **PPG debt service-to-exports ratio (in percent)** | 22.8 | 22.2 | 17.6 | | | 18.0 | 17.4 | 16.7 | 15.6 | 14.5 | 14.2 | | 2.5 | 0.8 | |
| Total gross financing need (billions of U.S. dollars) | 1.0 | 1.1 | 1.0 | | | 1.1 | 0.9 | 1.1 | 1.2 | 1.4 | 1.7 | | 1.9 | 2.0 | |
| Non-interest current account deficit that stabilizes debt ratio | -6.8 | -8.4 | 1.2 | | | -0.7 | -4.9 | -3.8 | -3.5 | -3.2 | -1.9 | | 6.4 | 5.2 | |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | -4.4 | -10.4 | -4.2 | -2.1 | 5.5 | -7.1 | -4.8 | -3.6 | -2.9 | -2.4 | -0.8 | -3.6 | 3.0 | 3.0 | 2.7 |
| GDP deflator in US dollar terms (change in percent) | 1.5 | 1.8 | 2.0 | 3.0 | 3.2 | 2.0 | 1.9 | 2.0 | 2.1 | 2.1 | 2.2 | 2.0 | 1.9 | 1.5 | 2.0 |
| Effective interest rate (percent) 5/ | 3.5 | 2.8 | 2.2 | 3.7 | 0.8 | 2.0 | 3.4 | 3.5 | 3.6 | 3.8 | 3.9 | 3.4 | 2.3 | 1.5 | 2.0 |
| Growth of exports of G&S (US dollar terms, in percent) | -14.8 | -8.1 | 7.6 | -1.1 | 11.0 | -2.1 | 6.2 | 4.0 | 5.6 | 5.4 | 5.9 | 4.2 | 5.0 | 5.0 | 5.0 |
| Growth of imports of G&S (US dollar terms, in percent) | -0.6 | -1.7 | 10.7 | 0.6 | 13.6 | 2.8 | -8.3 | 5.1 | 5.2 | 5.4 | 8.8 | 3.2 | 4.5 | 4.5 | 4.5 |
| Grant element of new public sector borrowing (in percent) | ... | ... | ... | ... | ... | 7.9 | 7.9 | 7.9 | 7.9 | 7.9 | 7.9 | 7.9 | 22.0 | 12.3 | 16.6 |
| *Memorandym item:* | | | | | | | | | | | | | | | |
| Nominal GDP (billions of US dollars) | 8.2 | 7.5 | 7.4 | | | 7.0 | 6.8 | 6.7 | 6.6 | 6.6 | 6.7 | | 8.5 | 13.2 | |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and $\rho$ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years.

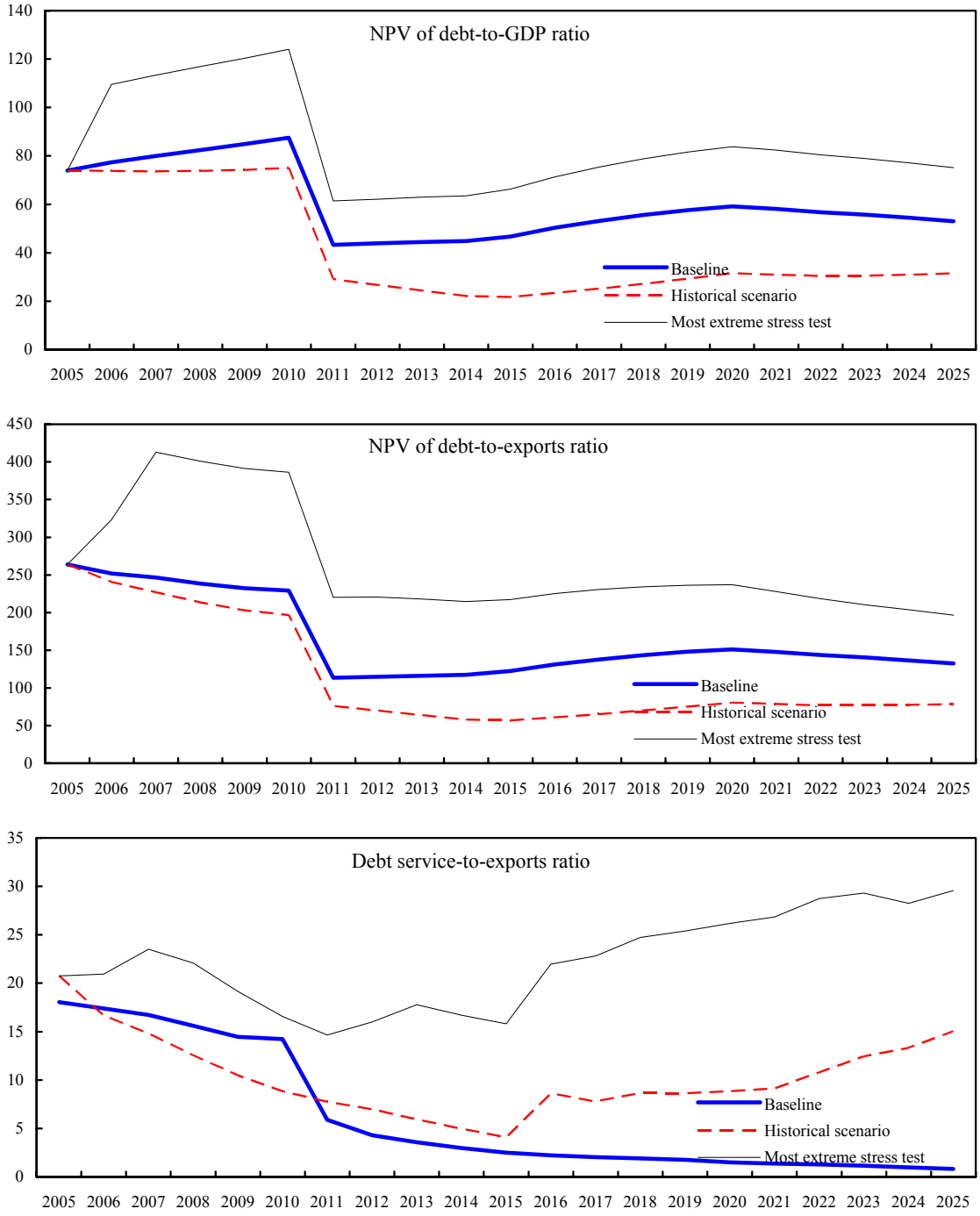
Table 1b. Zimbabwe: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

| | Estimate | Projections | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | **2015** | 2025 |
| **NPV of debt-to-GDP ratio** | | | | | | | | |
| **Baseline** | 74 | 77 | 80 | 82 | 85 | 88 | **47** | 53 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 1/ | 74 | 74 | 74 | 74 | 74 | 75 | **22** | 31 |
| A2. New public sector loans on less favorable terms in 2006-25 2/ | 74 | 77 | 80 | 82 | 85 | 88 | **54** | 76 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 74 | 80 | 86 | 89 | 91 | 94 | **50** | 57 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 3/ | 74 | 82 | 94 | 97 | 100 | 103 | **58** | 55 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 74 | 79 | 83 | 86 | 89 | 91 | **49** | 55 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/ | 74 | 79 | 83 | 86 | 89 | 91 | **50** | 54 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 74 | 82 | 94 | 97 | 100 | 103 | **58** | 56 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/ | 74 | 110 | 113 | 117 | 120 | 124 | **66** | 75 |
| **NPV of debt-to-exports ratio** | | | | | | | | |
| **Baseline** | 264 | 252 | 247 | 239 | 232 | 229 | **122** | 132 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 1/ | 264 | 241 | 227 | 214 | 203 | 197 | **57** | 79 |
| A2. New public sector loans on less favorable terms in 2006-25 2/ | 264 | 252 | 247 | 239 | 232 | 229 | **142** | 189 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 264 | 252 | 247 | 239 | 232 | 229 | **122** | 132 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 3/ | 264 | 323 | 413 | 401 | 391 | 386 | **217** | 197 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 264 | 252 | 247 | 239 | 232 | 229 | **122** | 132 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/ | 264 | 257 | 257 | 249 | 243 | 240 | **130** | 134 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 264 | 302 | 358 | 347 | 339 | 334 | **187** | 173 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/ | 264 | 252 | 247 | 239 | 232 | 229 | **122** | 132 |
| **Debt service ratio** | | | | | | | | |
| **Baseline** | 18 | 17 | 17 | 16 | 14 | 14 | **3** | 1 |
| **A. Alternative Scenarios** | | | | | | | | |
| A1. Key variables at their historical averages in 2006-25 1/ | 21 | 17 | 15 | 13 | 10 | 9 | **4** | 15 |
| A2. New public sector loans on less favorable terms in 2006-25 2/ | 21 | 17 | 16 | 14 | 12 | 10 | **9** | 23 |
| **B. Bound Tests** | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2006-07 | 21 | 17 | 16 | 14 | 12 | 10 | **7** | 20 |
| B2. Export value growth at historical average minus one standard deviation in 2006-07 3/ | 21 | 21 | 24 | 22 | 19 | 17 | **16** | 30 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07 | 21 | 17 | 16 | 14 | 12 | 10 | **7** | 20 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/ | 21 | 17 | 16 | 14 | 12 | 10 | **8** | 20 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 21 | 20 | 21 | 19 | 17 | 14 | **13** | 26 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/ | 21 | 17 | 16 | 14 | 12 | 10 | **7** | 20 |
| *Memorandum item:* | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 11 | 11 | 11 | 11 | 11 | 11 | **11** | 11 |

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 1. Zimbabwe: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005-2025
(In percent)




Source: Staff projections and simulations.